

Developing an online data-driven approach for prognostics and health management of lithium-ion batteries

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HIGHLIGHTS

- A novel data-driven prognostics and health management methodology is proposed.
- A health diagnosis method based on recurrent neural networks is proposed.
- A similarity-based model is developed for health prognostics.
- The state of health is estimated with a maximum root mean squared error of 0.61.
- Proposed model can predict the RUL with maximum absolute error of 110 cycles.

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ABSTRACT

Lithium-ion batteries have achieved dominance in energy storage systems. Meanwhile, there is a demand for the reliability of lithium-ion batteries. Battery prognostics and health management (PHM) is a discipline that not only provides accurate, early, and online health diagnosis, but also guarantees a robust and precise prediction of the remaining useful life of lithium-ion batteries, independent of the operating conditions. This paper attempts to develop a novel PHM methodology that addresses the points mentioned above. A large dataset including thirty-eight nickel manganese cobalt oxide battery cells is used. The battery cells have been tested under various test conditions to achieve different aging patterns. Afterward, the health indicators that describe the health trajectory of the battery are extracted from partial charging voltage curves. A recurrent neural network called nonlinear autoregressive with exogenous input is developed to estimate battery state of health (SOH) based on the extracted health indicators. The estimated SOH is used as the prognostic feature to develop a remaining useful life of battery (RUL) prediction model based on the similarity-based model. The proposed methods are validated using untrained data. The results indicate that the proposed PHM methodology can estimate the SOH of untrained battery cells with a maximum RMSE of 0.61. The RUL of battery cells with different aging patterns can be predicted with a maximum absolute error of 110 cycles. It can be concluded that the proposed method has the advantages of high precision in the health diagnosis and prognosis of battery cells regardless of their aging patterns, simplicity, and generalization to untrained data. These advantages point out the feasibility of the proposed method for online prognostics and health management of lithium-ion batteries.

Abbreviations: ANN, Artificial neural networks; BMS, Battery management system; BOL, Beginning of life; BP, Backpropagation; BR, Bayesian regularization; CC, Constant current; CV, Constant voltage; DOD, Depth of discharge; DTW, Dynamic time warping; ELM, Extreme learning machine; EOL, End of life; ESS, Energy storage systems; FEC, Full equivalent cycles; GPR, Gaussian process regression; HIs, Health indicators; HPPC, Hybrid pulse power; KPIs, Key performance indicators; LiFePO₄, Lithium iron phosphate; Li-ion, Lithium-ion; LM, Levenberg-Marquardt; LSTM, Long short-term memory; LTO, Lithium titanate; MSE, Mean squared error; ML, Machine learning; NARX, Nonlinear autoregressive with exogenous input; NMC, Nickel manganese cobalt oxide; OCV, Open circuit voltage; PHM, Prognostics and health management; RMSE, Root mean squared error; RNN, Recurrent neural networks; RPT, Reference performance tests; RUL, Remaining useful life; RVM, Relevant vector machine; SOC, State of charge; SOH, State of health; SVM, Support vector machine; TDLs, Time-delay lines.

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1. Introduction

1.1. Brief overview and research motivation

Lithium-ion batteries have been broadly used as energy storage systems (ESS) in electric mobility and stationary applications [65]. The assessment of battery life is a critical element in the success of ESS since the failure of the battery system imposes performance inefficiency, augmented degradation, and expensive maintenance [66]. Therefore, a reliable maintenance strategy should be pursued due to strict demands for safety in ESS and to satisfy the cost constraints [3]. Battery prognostics and health management (PHM) is an advanced solution that has emerged to optimize battery maintenance and ensure that the battery can satisfy its application's power and energy demand [4]. Battery health diagnosis and prognosis is not a novel field of research. Many researchers developed battery health diagnosis and prognosis models based on the NASA dataset. However, some points can be considered. First, the aging mechanism of Li-ion batteries highly depends on the factors such as battery chemistry, the stress factors, and the condition in which the battery operates. Most researchers select the limited number of cells from NASA dataset (Mostly B0005, B0006, B0007, and B0018). However, there is a low variety in the stress factors (temperature, DOD, rate of SOC,...) applied for cycling the batteries. Therefore, the battery cells have almost the same aging pattern. There have been significant developments in the field of battery PHM in the past decades. A comprehensive review of battery PHM methods is conducted in Ref. [5]. The objective of a PHM algorithm is to provide not only accurate, early, and online health diagnosis but also to predict the remaining useful life (RUL) of the lithium-ion batteries, with high precision, within a confidence interval, and independent of the operating conditions [6]. However, most of the previous researches has focused on some aspects of the PHM algorithm, and they have failed to cover all of these objectives in their research. For instance, a recent study proposed a data-driven prognostics and health management method based on voltage-temperature health indicators and the Gaussian process technique [7]. The authors in this study trained their model using extracted features from five battery cells. Two battery cells are used to validate the generalization of their method on untrained data. However, some points are neglected in their study. First, the battery cells used in their study are cycled under the same temperature (30 °C). Therefore, the proposed method in their study is limited to specific operating conditions. Moreover, the RUL prediction method proposed in their study cannot predict the RUL values at the early stages and steps-ahead of the end of life (EOL) of the battery cell.

Addressing all the shortcomings of previous studies has motivated the work in this paper to develop a comprehensive PHM methodology for lithium-ion batteries.

1.2. Background

Battery prognostics and health management (PHM) aims to determine the current and forecast the future state of the battery system based on real-time analysis of the significant parameters of the system and the subsequent actions [8]. Developing a PHM methodology reduces the critical failure and maintenance cost and increases the reliability of the battery systems [9]. Generally, battery PHM includes three critical components of condition monitoring and data acquisition, health diagnosis, and health prognosis [10]. Condition monitoring refers to monitor battery states and acquire the critical battery parameters. Various parameters can be considered in the process of battery PHM. However, voltage, current, temperature, and resistance are the most common parameters used by most researchers [11].

The next component is the health diagnosis. This component is one of the critical steps of the battery PHM process, and it concerns the state of health (SOH) estimation [12]. This component includes two steps of health indicators (HIs) extraction and developing a model for SOH

estimation. HIs are the features of a battery that display a predictable degradation trend through battery life trajectory [13]. Generally, there are two types of battery HIs, processed indirect and direct HIs [14]. The first type includes the features that are extracted from battery variables. This type of HIs is usually obtained by applying various techniques such as signal processing techniques, incremental capacity (IC), and differential voltage (DV) analysis. For example, Ref. [13] applied signal processing techniques to extract time and frequency domain HIs from the real-time voltage signal. The authors in Ref. [15] used the pattern of voltage and current signal as HIs. They applied the dynamic time warping technique (DTW) to compare the pattern of real-time battery signals at different stages of battery life. The value and position of the IC peak and valley show a distinctive pattern as the battery degrades. Therefore, they are considered as HIs by numerous researchers. For instance, Berecibar et al considered the detected peaks and valleys in IC and DV curves as HIs. The extracted three peaks form IC curve and two valleys and one peak from DV curve. The extracted HIs are then used as input of two ML methods to estimate the SOH [16]. Similarly, Yi et al [17] proposed IC peaks as a promising battery HI. They apply Gaussian filtering method to reduce the noise existing in IC curves and then extracted the peaks from smoothed curves. The authors in [18] proposed using the IC peak tracking approach for battery SOH estimation.

Similarly, the peak and valley positions of DV curves are widely used as HIs. The processed indirect HIs demand an extensive amount of data processing and filtering. Notably, extracting this type of HIs from battery parameters imposes an excessive computational burden [19]. Therefore, they are not convenient for real-time battery PHM. On the other hand, the direct HIs are recorded by the battery management system (BMS), and they can be directly obtained from its sensors. Voltage, current, and temperature are amongst the first type. The key advantage of this type of HIs is their applicability in online battery health monitoring. That is mainly because they are easily available from BMS during real-time operations [20]. Therefore, in this paper, direct HIs are used.

After selecting the appropriate HIs, a predictive model should be selected to estimate the SOH. Over the past decade, a variety of health diagnosis methods have been developed. A common strategy is to use the models characterizing the battery parameters and simulating its behavior [21]. These methods are called model-based methods [22]. The model-based methods focus on developing the electromechanical model [23], empirical model [24], or equivalent circuit model [25] of battery to analyze the degradation process and failure mechanism of lithium-ion batteries [22]. There exist an extensive range of literature that adopt the model-based methods along with a variety of observers and adaptive filters such as H-infinity filters [26], particle filters [27] and, Kalman filters [28] to estimate the current state of the battery. The model-based methods are precise. However, due to their demand for extensive computational effort and domain knowledge, there are growing appeals for data-driven methods. The data-driven methods, particularly machine learning (ML) models, have been influential in health diagnosis due to their low complexity and ability to be implemented in real-time applications [29]. A comprehensive review of the data-driven SOH estimation method is found in Ref. [30]. There are several common types of ML methods such as support vector machine (SVM) [31], relevant vector machine (RVM), Gaussian process regression (GPR), ensemble methods such as random forest regression and gradient, boosted trees, and different types of artificial neural networks (ANN) [32], have been widely adopted in the field of battery health diagnosis. The SVM method has been adopted by many researchers in battery health diagnosis [33]. It is a nonparametric method that can be applied to both linear and nonlinear data [34]. SVM approach requires small training samples, and it is robust to outliers. However, it has poor uncertainty, sparsity, and high computational cost [35]. Datong et al. proposed the RVM method for health prognostics and diagnostics of lithium batteries to address the disadvantages of the SVM approach [36]. Although the RVM model addresses some drawbacks of the SVM method, it requires extensive data for modeling. Moreover, it may fall

into a local optimization problem [13].

The GPR is a kernel-base, nonlinear and, nonparametric methodology that works based on Bayesian theory. Bayesian-based methods provide the uncertainty management associated with prediction, making them a suitable choice for high dimension and nonlinear data [37]. As a result, it has been adopted by many researchers for battery prognostics and diagnostics [38–39]. For instance, the authors in Ref. [40] proposed a GPR model to capture the dependency between the condition indicators extracted from voltage data and battery SOH. Notably, the GPR method has a high computational cost, and its performance highly depends on kernel function [6].

Ensemble learning methods are other types of ML methods that are employed for battery health diagnosis and prognosis. Li et al. [20] proposed random forest regression for SOH estimation. They used the features extracted from voltage-capacity data as input of the random forest model to estimate the SOH of NMC cells. The authors in [14] also developed a type ensemble learning method called gradient boosted tree (GBT) to estimate the SOH of NMC cells that were cycled under real driving profile. The ensemble learning methods are unlikely to overfit. However, they have learning time and memory constraints [6].

Among the methods mentioned above, various implementations of neural networks such as extreme learning machine (ELM), feed-forward neural network convolutional neural network (CNN), long short term memory (LSTM) approach, and recurrent neural network (RNN) prove their dominance in health diagnosis due to their precision, robustness, good performance on nonlinear data, and generalization to unseen data [41]. For instance, Pan et al. [42] applied the ELM method to estimate the battery SOH based on multiple health indicators. Kaur et al. [43] applied FNN, CNN, and LSTM methods for battery SOH estimation. They conclude LSTM approach outperforms two other methods in terms of accuracy.

Nonlinear autoregressive with exogenous input (NARX) is a type of recurrent neural network method that has achieved great success in many fields of research, such as real-time control of an unmanned aerial vehicle (UAV) [44], estimation of wind speed [45]. However, to the best of our knowledge, few studies have considered this method in the context of battery health assessment [46]. The authors in Ref. [46] adopt the NARX method for estimating the SOH of lithium iron phosphate (LiFePO_4) and lithium titanate (LTO) cells. To fill this gap, in our past work, we investigated the NARX method for the health diagnosis of nickel manganese cobalt oxide (NMC) battery cells [47]. In our past study, we have demonstrated the powerful ability of this method in the health diagnosis of twenty-one NMC cells that are cycled under various operating conditions. Therefore, herein, we applied this method for the health diagnosis step in the proposed PHM algorithm. The last component of the PHM algorithm is health prognosis. It is worth noting that almost all the methods applied for health diagnosis can be used for health prognosis. One of the well-known methodologies that are widely considered to predict the RUL value of a system is a similarity-based method. The similarity-based model predicts RUL only based on past cases with failure history, and thus it does not require any explicit failure criteria. What is more, it needs minimal prior knowledge of the degradation mechanisms [48]. In this method, a data repository is built containing the degradation patterns of reference components. Then, the similarity between health degradation trajectories of various reference components is computed. Finally, the RUL of a test component can be predicted based on the RUL of similar components [49]. This approach is particularly effective when a large dataset of a system is available, and it is broadly applied in many fields of researches such as bearing life prognosis [50], turbofan engine degradation [51] and, aero-engines [52]. Nevertheless, to the best of our knowledge, this method has rarely been studied in the field of battery RUL prediction. Therefore, it is of interest to investigate the performance of this method in the context of battery RUL prediction.

1.3. Contribution and novelty of the paper

It is notable that due to many unknown and unpredictable factors influencing the health of the battery, estimating battery SOH becomes quite challenging. A data-driven model which is developed based on the limited number of battery cells with a similar aging pattern suffers from lack of generalization capability to the battery cells with different chemistries and aging patterns. As the size of the available dataset increases, the predictive model will enhance in terms of prediction/generalization capabilities. Therefore, there are still some rooms to develop battery health diagnosis and prognosis for different battery chemistry, using a more comprehensive dataset. Some points make our research distinctive from other research in the field of data-driven battery prognostics and health management. The key contributions made to the field of battery PHM can be summarized as follow:

- 1) **Proposing a novel online data-driven PHM methodology:** This study proposes a comprehensive PHM methodology that includes data acquisition, health indicators extraction, health diagnosis, and health prognosis. The proposed PHM method is a straightforward method that can estimate the SOH accurately and predict the RUL steps ahead of the EOL of the battery.
- 2) **Using a comprehensive dataset:** Many previous studies focus on limited battery datasets. For instance, the recent studies [53–56] developed the data-driven health diagnosis and prognosis methods based on limited numbers of battery cells that are provided by NASA Ames Prognostics Center of Excellence. In contrast, this research uses a comprehensive battery dataset including thirty-eight NMC cells collected from two different comprehensive battery testing campaigns performed during several years. The proposed dataset covers a variety of stress factors that impact battery health degradation. This aspect of our work has a tremendous impact on the accuracy of the proposed battery PHM method.
- 3) **Proposing an accurate and reliable health diagnosis method:** As the data-driven methods rely on an extensive amount of data, the results achieved in our study would be more accurate and reliable compared to those that are developed based on the limited number of data. Moreover, the NARX method has a superior performance for nonlinear and complex data. As we will elaborate further, the results of our work show that the NARX method can estimate the SOH of the untrained cells that have different aging patterns from those used in training with minimum and maximum RMSE 0.14 and 0.61, respectively. In contrast, the results achieved by Ref. [53] show that the proposed data-driven method in their study can estimate the battery SOH with minimum and maximum RMSE of 0.77 and 1.16, respectively.
- 4) **Proposing the health indicators that can easily be obtained from BMS in real-time operation:** A battery health diagnosis and prognosis method should be straightforward. The models that include different data processing such as data filtering, data normalization, and signal processing are not suitable for implementing in BMS because they impose a computational burden. However, our model is a straightforward method that uses the data that are accessible during real-time operation. This is beneficial for applying the proposed method in real-time applications.
- 5) **Independency of proposed health prognostics method on battery operating conditions:** Normally, the performance of most developed models in this field depends on some specific operating conditions. They do not get generalized on other battery cells that operate under different conditions. That is because different operating conditions are neglected in the process of developing the health prognosis model. In contrast, 17 different operating conditions are considered in this research.
- 6) **Proposing the similarity-based method for battery RUL prediction:** To the best of the authors' knowledge, this is the first work that is proposed using the ensemble framework based on NARX and

similarity-based model for battery prognostics and health management. A similarity-based model is a non-parametric RUL prediction technique designed especially for engineering system. This method has the advantages of simplicity and low computational burden. Moreover, does not need any hypothesis to build the degradation model. Despite, the high precision, simplicity, and broad applicability of the similarity-based model in different fields of engineering, it has not been studied in the field of battery health prognosis. This is mainly due to the reason the developing a similarity-based model demands a comprehensive dataset which is not available for most researchers and generating such a data set takes some years. However, as a part of two European projects, our research team was successful to generate this dataset in more than two and a half years.

The remainder of the paper is organized as follows: Section 2 describes the framework of the proposed method for battery PHM. Section 3 to 5 describes each component of the PHM framework in detail, including data acquisition, health diagnosis, and health prognosis. The experimental results are discussed in Section 6. Section 7 presents a discussion and a comparative study. The challenges and future works are outlined in Section 8. Finally, the summary of the paper is presented in Section 9.

2. The framework of the proposed method

In this paper, a two-stage framework is proposed to develop an online PHM model for lithium-ion batteries. The proposed framework of the PHM methodology is depicted in Fig. 1. The offline phase includes the main components of data acquisition, health diagnosis, and health prognosis. In the data acquisition component, a comprehensive battery experiment was conducted on two types of NMC cells. The voltage and capacity data acquired from the battery experiment were maintained in a data repository for further analysis. These data are fed to the next component. The health diagnosis component includes two steps of health indicator extraction and developing an ML technique based on NARX for SOH estimation. The health indicators extracted from the partial voltage charging curve are used as the input of the NARX model

to estimate the SOH. The health prognostics component cannot work as a standalone component, and it uses the input from the health diagnosis component. In the health prognosis component, a repository of health degradation data is created. The repository includes the degradation patterns of the training data. Afterward, the similarity-based model is developed to predict the RUL of the battery.

The online phase includes online condition monitoring, health diagnosis, and health prognosis. During the online condition monitoring, the voltage data are recorded within a specific interval. Subsequently, the trained NARX model is used to estimate the current SOH of the battery. The estimated SOH is used as input data of the trained similarity-based model to predict the RUL of the battery. Finally, the predicted RUL is sent to the decision support in which the maintenance is scheduled. In the following subsections, each component of the PHM framework will be discussed in detail.

3. Data acquisition

Battery data acquisition is the process of collecting data from BMS sensors. Acquiring a substantial dataset is a prerequisite of a convenient battery PHM program. A comprehensive battery dataset should include a variety of battery cells with different degradation patterns. The battery experiment was performed on two groups of NMC cells, called group A and B, from two different manufacturers to achieve such a dataset. Group A includes twenty-one NMC cells. The experiment conducted on the cells in group A was part of the European Batteries2020 project, which took over two years and a half. All the test activities in group A were performed by using PEC manufactured ACT0550 battery cyclers.

The experimental work on group B was conducted for one and half year-long aging test campaign. A total of seventeen battery cells are considered for group B, which are cycled as part of an extensive test campaign in the battery innovation center (BIC) following an in-house test procedure. PEC manufactured CT0550G2 model battery cycler is used to perform cycle life tests and reference performance tests, and CTS made climate chambers are used to control the ambient test environment. The specifications of investigated cells of groups A and B are listed in Table 1.

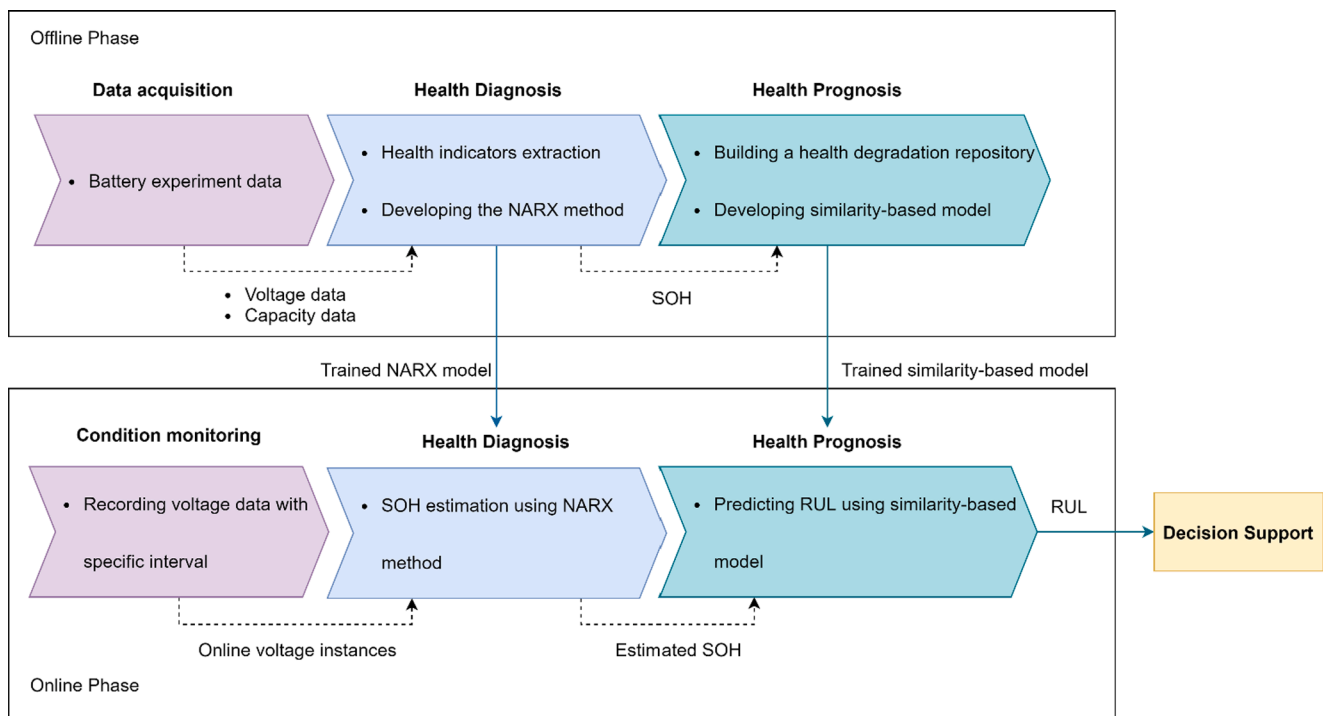


Fig. 1. The proposed framework of battery PHM. It includes online and offline phases.

Table 1

The specifications of the investigated battery cells.

Specifications	Group A	Group B
Cell technology	NMC/C	NMC/C
Rated capacity	20Ah	43Ah
Nominal voltage	3.65V	3.6V
Operating voltage	3–4.15V	3–4.2V
Internal resistance	$\leq 3.5\text{m}\Omega$	$\leq 2\text{m}\Omega$
Power density	2300W/Kg	$\geq 1200\text{W/kg}$
Dimension	7.1 mm × 217mm × 130mm	27.5mm × 148mm × 91mm
Weight	428g	840 g ± 10g

3.1. Experimental conditions

It is crucial to identify the aging path of fresh cells at different operating conditions to understand the process of battery health degradation. Then, multiple cycling test conditions are used to degrade the battery cells described in Table 2. The variations in the cycling tests include temperature (10, 25, 35, and 45 °C), depth of cycling (DoD) (20%, 50%, 60%, 80%, 90% and 100%), mid-SoC (50%, 65% and 80%), charge and discharge C-rates (0.33C, 1C, 2C, and 3C). Here, C-rate refers to the rated capacity of the cell.

3.2. Reference performance test (RPT)

Before the long-term investigation, the considered cells are pre-conditioned at the beginning of life (BOL) by reference performance tests (RPT). This type of test is performed to characterize the battery at BOL, cycling tests, and EOL to get the key performance indicators (KPIs). A typical RPT consists of three different characterizations: a capacity test, hybrid pulse power characterization (HPPC) test, and open-circuit voltage (OCV) test. All the performed tests are briefly explained below.

Preconditioning test: A couple of charge–discharge cycles is performed with a C/2 rate to activate the ions inside the cell. The preconditioning test can also confirm the rated capacity of the battery and a stable electrical connection.

Capacity test: Two types of capacity checks are performed at room temperature by cycling the battery thrice with a C/2 rate and C/5 rate, respectively. The last discharge capacity value is recorded as the cell's nominal capacity, which is used in the remaining useful life calculation.

Hybrid pulse power characterization (HPPC) test: The HPPC is performed at three different SoC levels of 20%, 50%, and 80%, with three different C-rates as 0.33C, 0.5C, and 1C. The internal resistance of the battery is calculated at 50% SoC with a 10 s 0.5C discharge pulse.

Cycle aging tests: The Li-ion cells are cycled following 17 different

aging conditions. The test matrix includes variations in operating temperature, DoD, mid-SoC, and C-rates. A full equivalent cycle (FEC) counter is used to count the performed cycle numbers. FEC is based on cycling charge Ampere-hour throughput, which allows comparing RPT results of different conditions.

When the cell reaches the specified 80% state of health (SOH), another RPT is performed at the EOL. The state of health of the battery is calculated as the following equation.

$$SOH = \frac{\text{actual capacity}}{\text{nominal capacity}} \times 100\% \quad (1)$$

4. Health diagnosis component

The health diagnosis component utilizes the voltage and capacity collected from battery experiments as input data. The output of this component is the estimated SOH at a given time. The health diagnosis component is already developed in our previous work [47]. In that work, we developed a data-driven health diagnosis method based on NARX. Herein, the proposed method is briefly discussed. The interested readers may refer to our previous work for more details.

4.1. Health indicators extraction

The first step of this component includes extracting the promising features that represent battery degradation behavior through battery life trajectory. As discussed in our previous work, the voltage charging curve acquired from the capacity test represents a predictable behavior as the battery degrades. Therefore, the features extracted from the voltage charging curve are convenient to develop a SOH estimation model. First, a voltage boundary with the value of 3.6 V for lower bound (V_l) and 3.8 V for the upper bound (V_u) was selected. Next, a sampling rate of 3mV was specified, and a window was introduced based on the specified sampling rate. The size of the window is calculated based on Eq. (2):

$$K = \frac{V_u - V_l}{\text{sampling rate}} + 1 \quad (2)$$

The feature extraction starts with recording the initial value of the window, which is V_l . Then the next feature is recorded when the initial value increases by the amount of sampling rate. The process of feature extraction will stop when the upper bound of the window (V_u) is recorded. The interested readers may refer to our previous work to understand the impact of different sampling rates and the voltage boundaries on our model.

Since each battery cell has been cycled for N full equivalent cycles (FECs), it includes N voltage charging curves. Bearing this in mind, the feature vector extracted from one cell is: $X = \{x_1, x_2, \dots, x_N\}$, where x is a

Table 2

The test conditions of the studied NMC cells.

Group	Test conditions	DOD %	Mid-SOC %	Temperature °C	Charge C-rate	Discharge C-rate	Number of cells
GROUP A	1	100	50	45	0.33C	1C	3
	2	100	50	35	0.33C	1C	3
	3	90	50	25	0.33C	1C	1
	4	80	50	45	0.33C	1C	2
	5	80	50	35	0.33C	1C	4
	6	50	65	35	0.33C	1C	3
	7	50	50	35	0.33C	1C	5
GROUP B	8	100	50	45	1C	1C	2
	9	100	50	35	1C	1C	2
	10	100	50	25	1C	1C	2
	11	100	50	25	3C	1C	1
	12	100	50	25	1C	3C	1
	13	100	50	10	1C	1C	2
	14	80	50	25	1C	1C	1
	15	80	50	25	1C	2C	1
	16	60	50	45	1C	1C	3
	17	20	80	45	1C	1C	2

feature vector extracted from one voltage charging curve.

4.2. Developing an ML method for SOH estimation

As explored in our previous work, the NARX has a powerful predictive capability. NARX is a type of recurrent neural network that estimates the value of $y(t)$ based on the d past values of $y(t)$ and $x(t)$. Considering $g(\cdot)$ as the function of nonlinear mapping, and d_x as input time delay d_y as the output time delays, the NARX is formulated according to Eq. (3).

$$y(t) = g(y(t-1), y(t-2), \dots, y(t-d_y), x(t-1), x(t-2), \dots, x(t-d_x)) \quad (3)$$

There are different selections of functions to train the NARX algorithm. Among them, Bayesian regularization (BR) and Levenberg-Marquardt (LM) are the most promising and well-known functions, which the former was selected in this work. The architecture of NARX is depicted in Fig. 2.

In NARX architecture, the time delay lines (TDLs) are defined according to Eq. (4).

$$TDL(t) = \begin{bmatrix} x(t-1), x(t-2), \dots, x(t-d_x) \\ y(t-1), y(t-2), \dots, y(t-d_y) \end{bmatrix}^T \quad (4)$$

The NARX algorithm can work in two different feedback modes that can be selected; series and series-parallel, which are depicted in Fig. 3.

In the series mode, the estimated output in the previous step is feedback to the input of the next step. However, if the real target output is available, the series-parallel mode provides higher accuracy and lower computational cost. Therefore, in this work, the series-parallel architecture mode is selected. The estimation accuracy of the proposed method is evaluated using the following metrics: root mean squared error (RMSE), mean squared error (MSE), and R-squared R^2 which are defined according to the following equations where N is the number of data points, $\hat{Y}_i$ and Y_i represent the estimated and real output respectively, and $\bar{Y}$ is the mean observed Y_i . Closer the value of MSE and RMSE to zero more accurate the model is. On the other hand, the R^2 value close to 1 demonstrates the accuracy of the model.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{Y}_i - Y_i)^2} \quad (5)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (\hat{Y}_i - Y_i)^2 \quad (6)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{Y}_i - Y_i)^2}{\sum_{i=1}^N (\bar{Y} - Y_i)^2} \quad (7)$$

5. Health prognosis component

Health prognosis is the fundamental component in the PHM framework that presents vital information about the battery RUL. RUL of a lithium-ion battery is the remaining number of cycles until it reaches the predefined failure threshold. The failure threshold of a battery cell in automotive applications is when the SOH reaches 80% of its initial value. The health prognostics component cannot work as a standalone component, and it should receive input from the health diagnosis component [6]. The SOH values estimated in the health diagnosis component are used as the prognostic features to develop an RUL prediction model. Herein, a similarity-based approach is proposed for the health prognostics of lithium-ion batteries. The main idea of similarity-based methods is that if an operating component at a specific time interval has a similar performance to a reference component, they present a similar degradation pattern. Therefore, the RUL of a battery cell can be predicted based on the similarity between its degradation history and the RUL of the battery cells with similar degradation patterns.

To develop a similarity-based model, a repository of degradation profiles should be available. Degradation profiles represent the evolution of the health indicators as a battery cell transitions from BOL to EOL. Degradation profiles are also called reference components. The evolution of battery SOH from BOL to EOL of life is required to build such a repository. However, some of the battery cells in our experiment did not reach EOL due to time limitations. In order to acquire the trajectory of SOH of a battery cell until the EOL, a model is required to describe the degradation behavior of each battery cell. Based on previous research, the behavior model can be obtained using curve-fitting techniques on the available data [18,57–58]. After creating a fit and obtaining the degradation behavior of a battery cell, the extrapolation method is used to estimate the SOH values outside the range of available data. Data will extend until the SOH reaches 80% of its initial value. The curve fitting methods will be explained in the discussion section.

In the next step, the pairwise similarity between health degradation trajectories of various reference components is computed. The RUL of a test component is calculated based on the median life span of the most similar reference components minus the current lifetime value of the test component. In this study, the dynamic time warping method (DTW) calculates the distance between the test and reference components.

DTW is the algorithm for computing the similarity between two degradation patterns that may have varying lengths. Considering two degradation profiles $M = \{m_i, i = 1 \text{ to } |M|\}$ and $N = \{n_j, j = 1 \text{ to } |N|\}$ where

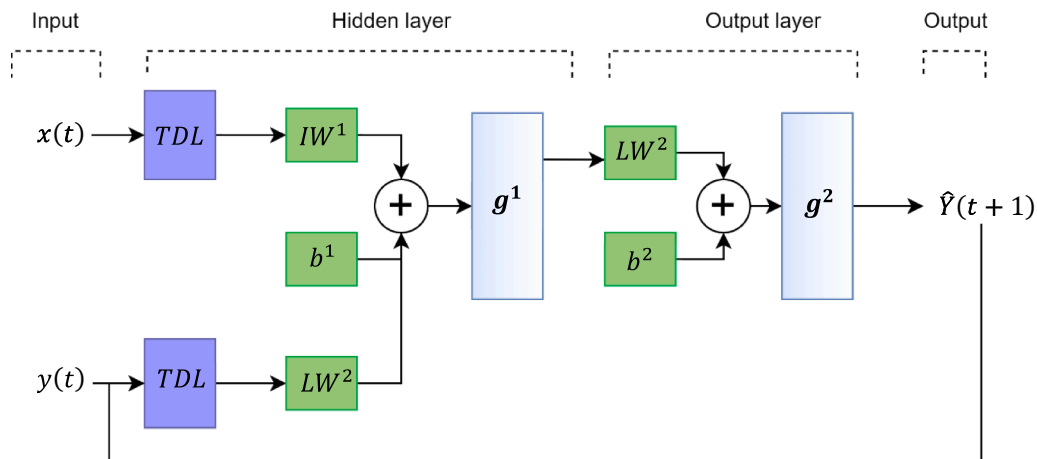


Fig. 2. The architecture of NARX. The b symbol represents the bias and IW, LW are the input weight and, layer weight for former and latter [47].

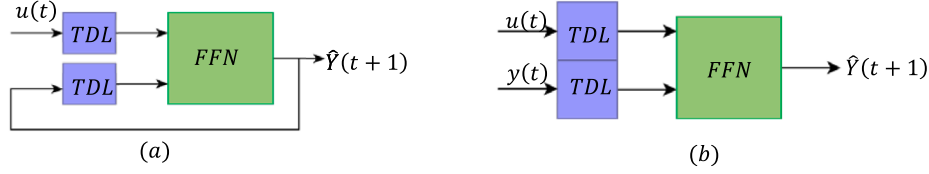


Fig. 3. NARX modes. a) Parallel mode, b) Series-Parallel mode [47].

$|M|$ and $|N|$ are the length of profiles M and N respectively. First a $|M| \times |N|$ distance matrix D is created. The value of D_{ij} is calculated based on the Euclidean distance between the element m_i and n_j according to Eq. (8).

$$D_{ij} = (m_i - n_j)^2 \quad (8)$$

To find the optimal similarities between the degradation patterns M and N , a warping path WP should be defined through matrix D that minimizes the entire cumulative distance between them. Considering the following assumptions: $WP_{0,0} = 0$, $WP_{i,0} = \infty$, and $WP_{0,j} = \infty$, the WP is defined based on the dynamic programming in Eq. (9).

$$WP_{ij} = \min\{WP_{i-1,j-1}, WP_{i-1,j}, WP_{i,j-1}\} + D_{ij} \quad (9)$$

Then the DTW distance between degradation profiles M and N is calculated as Eq. (10).

$$DTW = WP_{|M|,|N|} \quad (10)$$

Finally, the accuracy of the proposed method in RUL prediction is evaluated based on the absolute error criteria.

$$Absolute\ error = |RUL_{real} - RUL_{predicted}| \quad (11)$$

6. Experimental results

6.1. Results of battery experiment

This section highlights the significant findings of the battery experiment. The aging mechanisms of battery cells could be divided into 1. those originated from the cell chemistry, majorly coming from the characteristics of the electroactive materials, and 2. those originated from the operating conditions. Considering the fact that the tested cells had the same electrode chemistry (NMC), it is expected that the obtained aging pattern has not been impacted by the issues that arose from the differences in electro-active materials. On the other side, the operational conditions which affect the aging process were repeatedly reported as

DoD, SoC, C-rate, and temperature. Having paid attention to this important point, our battery cells were tested under various aging conditions (Table 2) covering diversified DoD, SoC, C-rate, and temperature to achieve an inclusive aging pattern. Fig. 4 illustrates the health degradation patterns of group A and group B.

As Fig. 4 demonstrates, the aging rate of battery cells affected by different stress factors such as temperature, DoD and SoC range, and charge-discharge rate. Below, the impact of each stress factor is discussed.

Temperature: Temperature has a tremendous effect in accelerating battery cell degradation. By comparing the aging rate of cell 1 and cell 4, in group A, that are cycled under the same conditions but at different temperatures, it is clear that the degradation rate increases at elevated temperatures. This also can be understood by comparing the aging patterns of cells 1 and 3 in group B. Moreover, the comparison between cell 5 and cell 9 in group B confirms that by decreasing the temperature lower than 25 °C, the battery cells tend to degrade faster. This is also in line with the findings of the previous study [59].

DOD: DOD is another critical factor in battery degradation. The higher range of DOD accelerates battery health degradation. For example, the cells that are cycled under test condition (4) have a higher rate of aging than those that are cycled under test condition (7). As illustrated in Fig. 4, cell 4 in Group A, reaches EOL after cycling for 12 FEC rounds while cell 17, which is cycled with the lower DOD, only reaches 95% of SOH after cycling for 15 FEC rounds. Comparing the life trajectories of cells 8 and 16 in group B also confirms the effect of a high range of DOD in increasing the degradation rate.

Range of SOC: In this work, there is low variety in the range of SOC due to practical reasons. The influence of the range of SOC can only be comprehended by a comparison of degradation patterns of cells cycled under test conditions (6) and (7). As Fig. 4 suggests, the range of SOC has a modest impact on battery health degradation. Nevertheless, the aging rates of cells 14 and 17, group A, indicate that the lower range of SOC to some extent hinders the process of battery aging.

Rate of Charge-discharge: Comparing the life trajectories of cells tested under test conditions 14 and 15 confirms that the increment of the

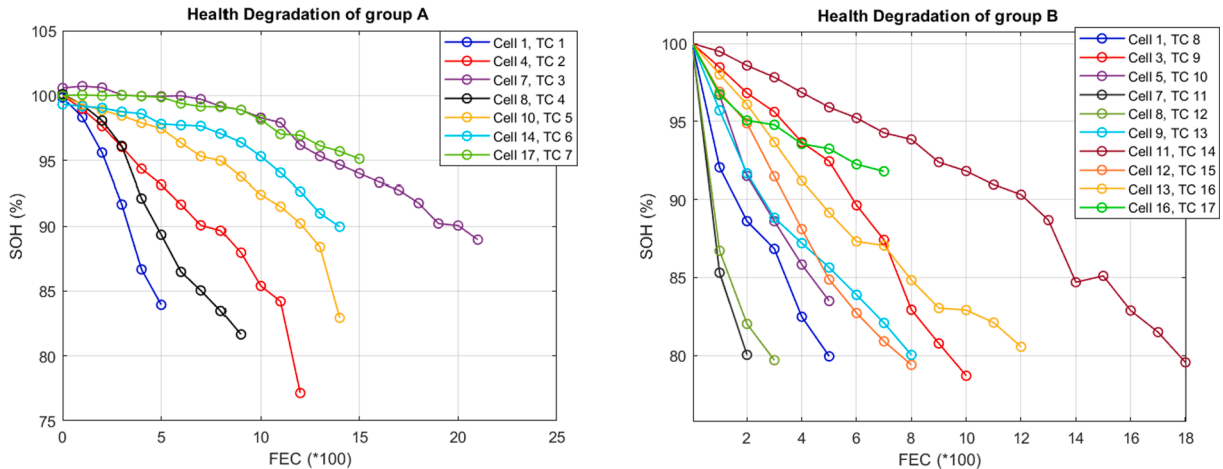


Fig. 4. Health degradation of cells in groups A and B. To enhance the figure comprehensibility, from each test condition (TC) presented in Table 2, one cell is selected to be presented in the figure. Cycling the battery cells under various stress factors, their degradation patterns are different.

charge–discharge rate leads to speeding up the aging process of lithium-ion batteries. As Fig. 4 illustrates, cell 11 has a longer lifetime than cell 12 that cycled under a higher discharge rate.

6.2. Results of health diagnosis component

This section discusses the result of health diagnosis using the proposed method. In this work, the NARX model is trained using the health indicators extracted from the voltage charging curves of the battery cells in group A and group B. Fig. 5 illustrates the evolution of voltage charging curves through the battery trajectory.

As can be seen, a voltage charging curve has a lower slope at BOL, whereas it presents a sharper slope when it reaches EOL. Moreover, the initial, mean, and last values of curves are different at various stages of battery life. Therefore, the features extracted from the voltage charging curve are convenient to develop a SOH estimation model. In the next step, from each test condition, Table 2, one cell is used for model validation, and the rest are used for training the model. It should be mentioned that, for the test conditions that consist of only one cell (test conditions (3), 11, 12, 14, and 15), the cells that are cycled under test conditions 11 and 14 were used for training. Those that are cycled under test conditions (3), 12, and 15 are used for validation purposes. Therefore, the proposed model was trained using 23 battery cells, and 15 cells were selected to validate the model accuracy for untrained data. The NARX network configuration is set as follows: The number of hidden neurons was considered as 10. The higher number of hidden neurons increases the estimation accuracy.

Nevertheless, it may cause overfitting. The number of hidden neurons is selected based on the trial and error approach to balance the estimation error and avoid the overfitting problem. The series-parallel mode and Bayesian backpropagation function were selected for training the network. Finally, the size of input and feedback delays is considered as 2. The entire training data set is divided into training, validation and testing subset during the training phase. 70% of the dataset is considered training data, and the rest of the data are used to validate and test the network. The network is trained with training data. Validation data are used to assess the generalization of the network and to terminate training when the generalization process stops improving. The testing samples evaluate network performance through and after training. Fig. 6 visualizes the error histogram between the real SOH and the estimated SOH in the training phase. Fig. 6 illustrates that approximately all the SOH estimation errors are spread around the zero-line error. This

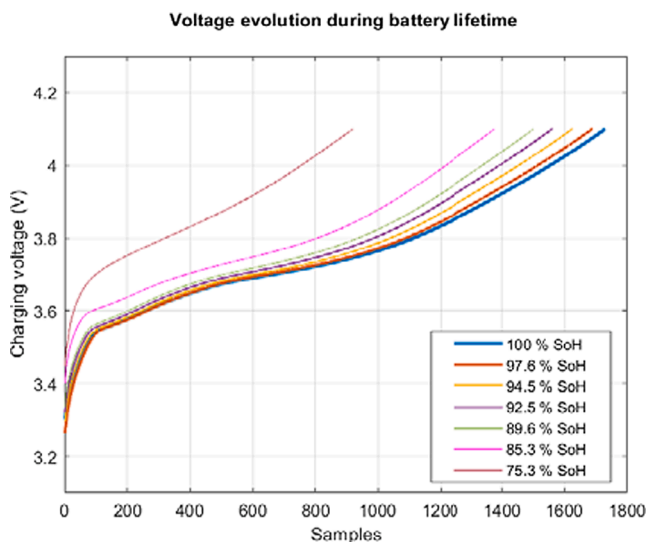


Fig. 5. Voltage evolution through battery lifetime, showing a distinct pattern as the battery reaches EOL.

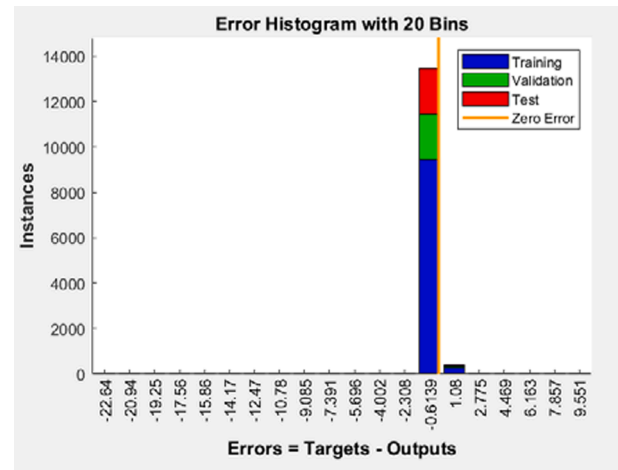


Fig. 6. Error histogram of NARX method in the training phase. All the SOH estimation errors are spread around the zero-line error. This indicates the accuracy of the model.

implies that the NARX has the ability of SOH estimation with high precision.

After training The NARX model, it is crucial to evaluate the capability of the trained model on the health diagnosis of unseen data. As mentioned before, the 15 battery cells are selected to validate the accuracy and generalization of the trained model. The health diagnosis results of battery cells in groups A and B are demonstrated in Table 3.

As results show, the proposed model provides the R-squared value of 0.99 for all validation samples. This means that the NARX model precisely fits the untrained data. Moreover, the low RMSE and MSE metrics values indicate that the difference between the real SOH and estimated SOH is small. In conclusion, the results prove that the NARX can estimate the SOH of untrained data with high precision. However, one can understand that the NARX has better performance in predicting the cells in group A. This may be because each test condition in group A includes more battery cells. Therefore, more data sample is used for training the model. It is also notable that the highest estimation errors belong to the cells cycled under test conditions 15. This is mainly because it did not participate in the training phase.

For the sake of illustration from each group, the cells with the highest estimation error are selected.

Fig. 7 (a) and (b) include two subfigures. The top subfigure shows the actual vs predicted SOH (the blue and red figure respectively), and the bottom subfigure indicates the residual of NARX in estimating SOH values. The residual error represents the difference between actual and estimated values. By analyzing Fig. 7 one can understand that the real

Table 3

The health diagnosis results of the validation dataset.

Group	Test condition	Cell No	RMSE	MSE	R ²
Group A	1	1	0.35	0.12	0.99
	2	4	0.31	0.10	0.99
	3	7	0.57	0.32	0.99
	4	8	0.38	0.18	0.99
	5	10	0.21	0.04	0.99
	6	14	0.25	0.06	0.99
	7	17	0.14	0.02	0.99
Group B	8	1	0.42	0.17	0.99
	9	3	0.45	0.19	0.99
	10	5	0.39	0.29	0.99
	12	8	0.54	0.29	0.99
	13	9	0.42	0.18	0.99
	15	12	0.61	0.38	0.99
	16	13	0.43	0.27	0.99
	17	16	0.39	0.18	0.99

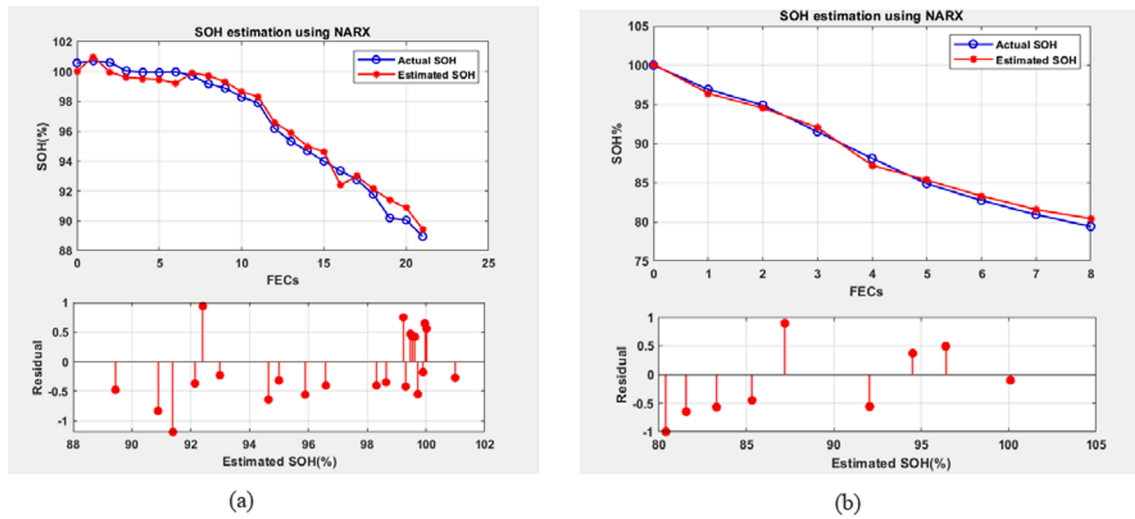


Fig. 7. Illustration of SOH estimation results of the cell cycled under the (a) test condition (3), and (b) test condition 15. As can be seen, the estimated SOH is closely matched with the actual results.

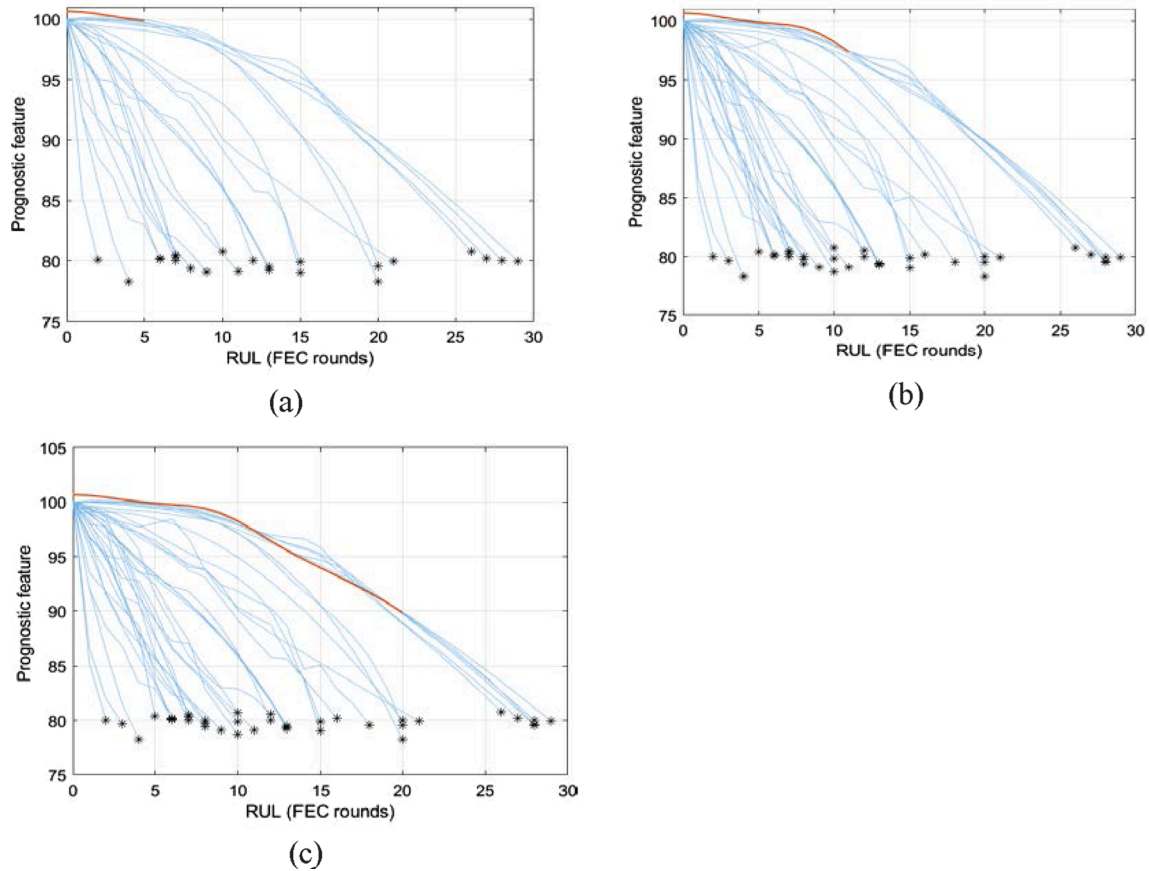


Fig. 8. Process of RUL prediction of a test component using similarity-based model a) using 20% of data b) using 40% of data c) using 70% of data. As Fig. 8 shows, to predict the RUL using the proposed method, first, a repository of degradation profiles of reference data is built (blue curves). Then at each iteration of the RUL prediction process, given the health trajectory of the cell, the similarity-based model finds the K numbers of reference components that are most similar to the test component (red curve). The RUL of the test component is calculated based on the EOL of most similar reference components (star marks). In other words, the model fits a probability distribution based on the nearby components, and the median of the distribution is considered as the prediction of RUL. When the tested cell is at the first stages of its life, more reference components are close to it. However, the reference curves bifurcate as the cell continues to progress its life trajectory. Finally, when the cell approaches EOL, the number of nearby reference components limits to few cells. Table 4 summarizes the results of RUL prediction on the validation (untrained) dataset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

SOH of an untrained cell can be estimated using NARX with high accuracy and low residual error. This confirms that the NARX has a high ability to generalization on unseen data. This makes the NARX model a convenient choice for the health diagnosis method in the framework of battery PHM.

6.3. Health prognosis results

This section summarizes the results achieved from the health prognosis component. As mentioned, the objective of the health prognosis component is predicting the RUL of the battery, steps ahead of the EOL of the battery, and independent of the operating conditions.

The same dataset that is used for training and validating the NARX model is used for training and validating the similarity-based model, i.e. 23 cells are used for training (reference components) and 15 cells are used for validating (test components) the model. For each test component in the validation dataset, 20%, 40%, and 70% of its data are fitted to the trained model to evaluate the effectiveness of the proposed method in the RUL prediction of the test components at different stages of their life trajectory. The process of RUL prediction for a test component (cell 3 from group A) using the similarity-based model is depicted in Fig. 8.

The results in Table 4 show a relatively large error between the predicted RUL and actual RUL when a battery cell is at the first stages of its life trajectory. However, when more data are observed and the cells approach EOL, the accuracy of RUL prediction is enhanced. To visualize the results, the actual RUL and the probability distribution of the predicted RUL of the cell with the highest prediction error (cell 3 from group A) are depicted in Fig. 9.

The black and the red vertical lines presented in Fig. 9 indicate the predicted RUL and the true RUL, respectively. The blue curve shows the probability density function of the predicted RUL. It is notable that as the life trajectory of the battery cells passes and more data are available, the predicted RUL approach to the true RUL. As can be observed, when more data are available, the model predicts the RUL more precisely, and the distribution becomes more concentrated. Finally, when 70% of health trajectory data are available and the battery cell is close to EOL, the proposed approach can predict the RUL with an absolute error of 1.1 FECs (110 cycles). This demonstrates that the proposed approach can predict the EOL of untrained battery cells with relatively high accuracy, steps ahead of their EOL. The results also highlight that another promising aspect of the proposed method is the ability to predict the RUL of the cells that are cycled under different test conditions from those used in training. This can be understood from the results achieved from the cells cycled under the test conditions that are not trained (test conditions (3), 12, and 15). Together, the present finding confirms that the proposed approach is suitable for the health prognostics of lithium-ion

batteries.

7. Discussion

7.1. Remarks

Battery PHM is a critical task to ensure the safety and reliability of the application they operate. A promising battery PHM methodology has high precision and low computational cost and should apply to battery cells with various aging behaviors. Moreover, it is evident that the proposed method can be practically applied in BMS. This research attempts to address the points mentioned above by developing a comprehensive PHM methodology. A comprehensive battery experiment is conducted on two types of NMC cells (group A and B) in this research. The degradation results are demonstrated in Fig. 4, and the impact of each stress factors discussed separately. It is shown that the cells cycled at elevated temperatures with a higher rate of DOD tend to degrade faster. Afterward, it is illustrated in Table 3 and Fig. 7 how well the NARX method performs on SOH estimation of unseen data. It was shown that the NARX performs better on SOH estimation of the battery cells in group A and the maximum RMSE (0.61) belongs to cell No.12 from group B. Lastly, the RUL prediction results are represented in Table 4. It is indicated that the similarity-based method has poor performance when the battery is at the first stages of its life. However, it can predict the RUL more accurately and between the confidence interval when sufficient data become available.

Meanwhile, there following points should be discussed. 1) The process of selecting the appropriate curve fitting method, 2) The superiority of the proposed method for battery health prognostics over another popular data-driven method, and 3) The applicability of the proposed PHM method in the BMS system. The following sub-sections are devoted to elaborate on the points mentioned above.

7.2. Comparison of different curve fitting methods

Unfortunately, there are no specific criteria to define which curve fitting function can describe the behavior of data more efficiently [60]. In this study, to find the promising fitting function, MATLAB Curve Fitting Toolbox was used. The toolbox lets the users apply various fitting functions and compare the result of candidate models. The most convenient fitting function will be selected by comparing numerical fit results, including the fitted coefficients and the goodness-of-fit. The goodness-of-fit defines how well the curve fits the data and, the confidence bounds on the coefficients define their accuracy. Herein, four popular fitting functions were nominated as candidates for fitting functions. The functions are Gaussian, exponential, Smoothing-spline and, polynomial. The numerical results of the candidate functions are

Table 4
RUL prediction results of validation dataset.

Group	TC	Cell No	Actual RUL (FECs)			Predicted RUL (FECs)			Absolute Error (FECs)		
			20%	40%	70%	20%	40%	70%	20%	40%	70%
Group A	1	1	5	4	2	8.5	7	2.1	3.5	3	0.1
	2	4	11	8	4	8.6	6.7	3.5	2.4	1.3	0.5
	3	7	23	17	8	18.6	13.9	6.9	4.4	3.1	1.1
	4	8	8	6	3	11.3	4.4	2.8	3.3	1.6	0.2
	5	10	13	10	5	14.9	9	4.6	1.9	1	0.4
	6	14	16	12	6	12.2	10.2	5.7	3.8	1.8	0.3
	7	17	23	17	8	19.5	14.4	7.1	3.5	2.6	0.9
Group B	8	1	4	3	1	8.1	7.3	1.7	4.1	3.7	0.7
	9	3	8	6	3	6.2	5.4	2.8	1.8	0.6	0.2
	10	5	7	5	2	6.3	5.5	2.2	0.7	0.5	0.2
	12	8	3	2	1	8.6	3.8	2	5.6	1.8	1
	13	9	7	5	2	6.1	4.6	2.2	0.9	0.4	0.2
	15	12	7	5	2	5.2	4.1	2.1	1.8	0.9	0.1
	16	13	10	7	3	6.9	6.7	2.7	3.1	0.3	0.3
	17	16	10	7	3	6.4	6.7	2.6	3.4	0.6	0.4

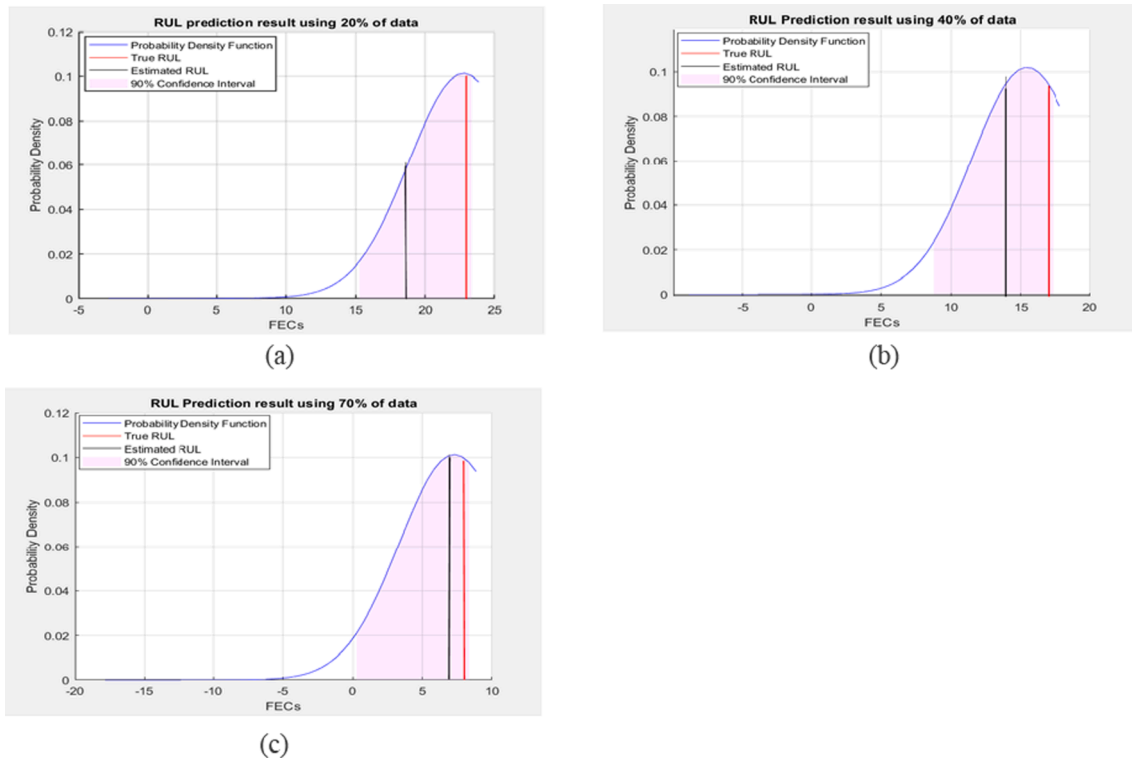


Fig. 9. Probability distribution plot at different life stages of the test component a) 20% of data b) 40% data c) 70% of data.

Table 5

Numerical results of curve fitting using the candidate functions. From each test condition, one cell was selected as input to compare the curve fitting functions. As the battery cells in the same test conditions present similar behavior, we skip examining all the battery cells.

Cell number	Gaussian		Exponential		Polynomial		Smoothing-spline	
	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
TC 1	0.98	0.80	0.97	1.08	0.98	0.85	0.97	0.96
TC 2	0.99	0.39	0.99	0.40	0.99	0.41	0.99	0.40
TC 3	0.99	0.36	0.99	0.38	0.98	0.41	0.98	0.40
TC 4	0.99	0.38	0.98	0.88	0.97	1.09	0.83	0.98
TC 5	0.98	0.53	0.94	1.06	0.95	0.99	0.98	0.64
TC 6	0.99	0.16	0.99	0.28	0.35	0.98	0.99	0.21
TC 7	0.99	0.14	0.98	0.23	0.98	0.21	0.98	0.22

compared to determine the best fit for our dataset. The results of curve fitting for cells in group A are depicted in Table 5. From each test condition, one cell was selected as input to compare the curve fitting functions.

By comparing the curve fitting results, one can understand that the Gaussian method provides lower error and higher performance compared to other methods. The results reveal that for the current dataset, the Gaussian function provides the best numerical results. To validate the effectiveness of the selected function in estimating the point that a battery cell reaches EOL, we use the data of battery cells that already reached EOL. For instance, cell 11 from group B, was selected as input data. This cell reaches EOL after 18 FEC rounds. 50% of the SOH data of this cell was fitted to the model to validate the accuracy of the fit model. The results are shown in Fig. 10.

In Fig. 10, the fit function 1 to four (Fit 1 to Fit 4) indicates the Gaussian, Exponential Smoothing-spline, and polynomial fitting functions, respectively. By analyzing the results, one can understand that the cell reach to EOL is estimated to be 25.4 FEC and 22.39, 19.3 FEC by the exponential and polynomial, and smoothing-spline functions. As is evident, the difference between the true value (18 FEC) and estimated values is great. Whereas the Gaussian model (Fit 1) precisely predicts the point that the SOH reaches 80%.

By analyzing the results from Table 5 and Fig. 10, it is concluded that the Gaussian function can be a convenient choice for our purpose. Therefore, the Gaussian fitting function is selected to capture the degradation behavior of battery cells.

7.3. The comparison of the similarity-based model with long short-term memory network

In this section, the proposed method for health prognostic is compared with the long short-term memory (LSTM) network. LSTM is a type of recurrent neural network that learns long-term dependencies between time steps of sequence data, and it is one of the well-studied methods for RUL prediction of lithium-ion batteries [61–63]. Fig. 11 demonstrates the architecture of an LSTM network for predicting the time series data. As shown in Fig. 11, the LSTM network for RUL prediction includes a layered array, an LSTM, a fully connected layer, and a regression output layer. The size of the input and fully connected layers was set to the number of features in input data and the number of responses, respectively. The main elements of the LSTM layer are “memory cell” and the gates, including the input gate (I_g), output gate (O_g), and forget gate (F_g). The memory cell controls which information should

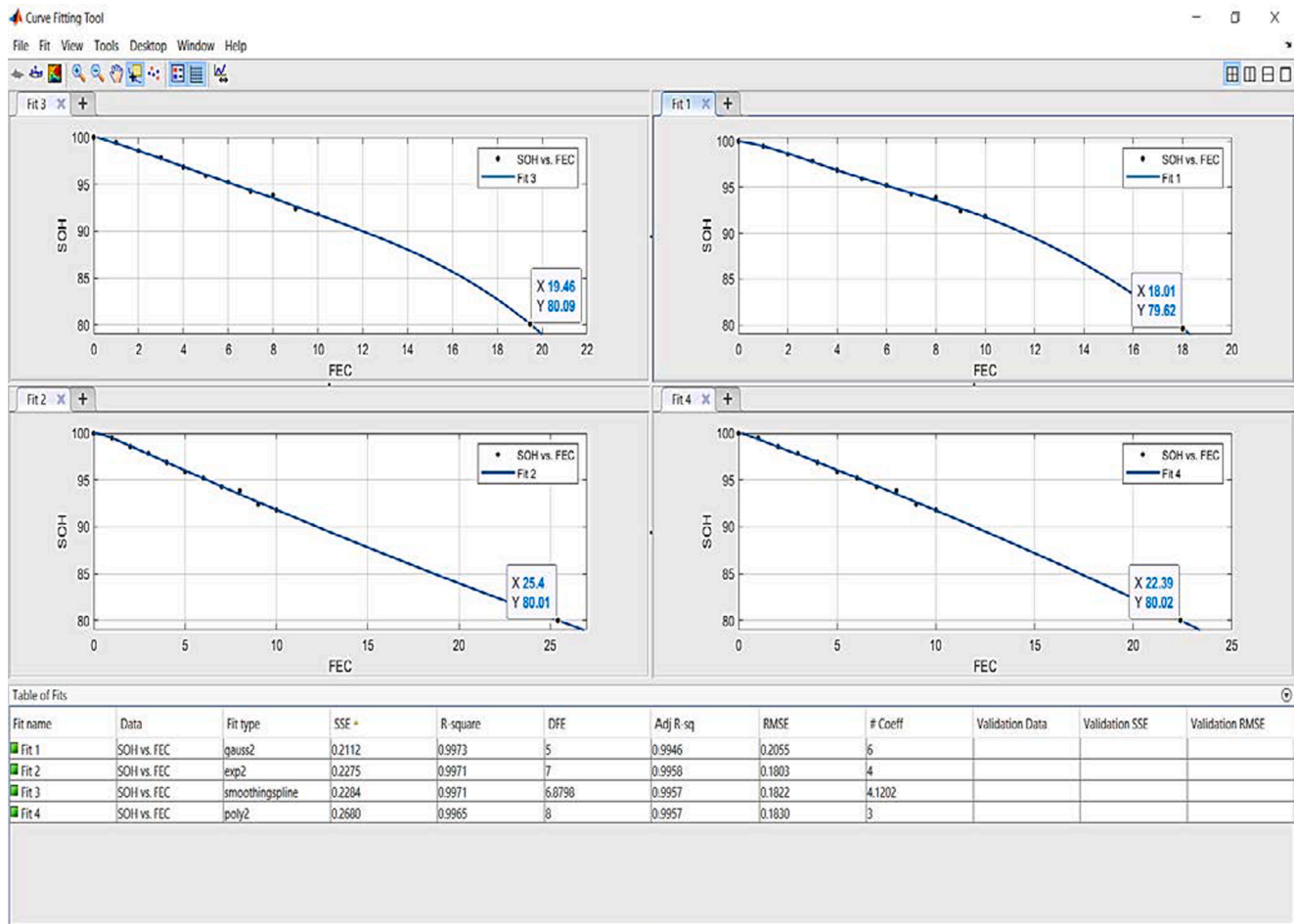


Fig. 10. The screenshot of the curve fitting toolbox in the MATLAB environment. It presents the graphical and numerical results of fitting cell 11 (group B) into four fitting functions. The goal of fitting SOH data is to extrapolate the best fit to predict that SOH reaches 80%.

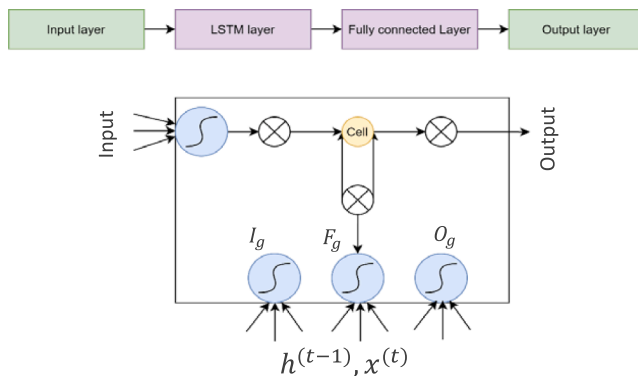


Fig. 11. The architecture of the LSTM network.

be added or should be removed. The LSTM layer uses two activation functions, one for the gates and the other for memory cell. The gates are the sigmoidal units that are activated by the input layer and the hidden layer of the previous time step (t). Another activation function is used to update the memory cell. The interested readers can refer to Ref. [64] for more information on the LSTM network.

The following configuration is adapted to train the LSTM network; the number of features (SOH) and the number of responses (RUL) equals 1. For the LSTM layer, the number of hidden neurons is specified to 125. The hyperbolic tangent function (Tanh) and the Sigmoid function are selected as the activation function for the memory cell unit and the

gates, respectively. Furthermore, the model is trained for 250 epochs. The LSTM network is trained using the same dataset that is used for training the similarity-based model. Table 6 summarizes the comparison results between the LSTM network and the proposed model. It is worth noting that the models were compared using 70% of validation cell data.

By comparing the results achieved from the proposed model and the LSTM network, one can understand that although in some cases the LSTM results are better, overall, the proposed method has a lower

Table 6

The comparative results between the proposed method and the LSTM network. These results are achieved by fitting 70% of the data of each validation cell to the trained network.

Group	TC	Cell No	The absolute error of the proposed model (FECs)	The absolute error of the LSTM network (FECs)
Group A	1	1	0.1	0.5
	2	4	0.5	0.3
	3	7	1.1	1.8
	4	8	0.2	0.6
	5	10	0.4	0.9
	6	14	0.3	0.4
	7	17	0.9	0.8
Group B	8	1	0.7	0.6
	9	3	0.2	0.7
	10	5	0.2	0.5
	12	8	1	1.4
	13	9	0.2	0.2
	15	12	0.1	0.4
	16	13	0.3	0.2
	17	16	0.4	0.6

absolute error in predicting battery RUL. This indicates the superiority of the proposed LSTM network method in RUL.

Regarding the computational burden, although deep neural networks have achieved effective results for battery health prediction, they have high computational complexity. Besides, their performances are highly dependent on selecting the appropriate parameters and architecture in the neural network. To deal with this issue, some researchers have proposed several *meta*-heuristic, including artificial bee colony (ABC), genetic algorithm (GA), particle swarm optimization (PSO). However, these methods impose more computational burden which is not suitable for real-time operation. On the other hand, the similarity-based model is easy to implement, has low computational time, and does not need any hypothesis to build the degradation model. In real-time operation, the proposed method can predict the RUL steps ahead of the EOL battery only based on the estimated SOH degradation pattern from BOL until the given time.

7.4. BMS applicability

In this section, the proposed PHM methodology is investigated from the BMS applicability viewpoint. As mentioned before, the workflow of the proposed PHM methodology in the online phase is as follows. The voltage data are recorded within a specific interval. Subsequently, the trained NARX model is used to estimate the current SOH of the battery. The estimated SOH is used as input data of the trained similarity-based model to predict the RUL of the battery. The proposed PHM methodology has a protentional to be implemented in a BMS system for the following reasons. First, we propose the health indicators that are easily accessible from BMS sensors during the real-time operation.

Moreover, the computational cost can be reduced by eliminating data pre-processing such as data smoothing or noise reduction from PHM workflow. This enhances the applicability of the proposed method in real-time application. From the perspective of accuracy and generalization of proposed models, it has been shown that both NARX and similarity-based models have high precision, and they are able to health diagnosis and prognosis of unseen data. Regarding the computational cost, the average time for health diagnosis and prognosis of an untrained cell is about 2.12 s which is appropriate for online health diagnosis and prognosis. The computational cost for one untrained cell includes the time spent on extracting the health indicators, SOH estimation, and RUL prediction using the pre-trained models. It is worth noting that the curve fitting method was skipped in computing the computational costs since in the online phase, there is no need for using the curve fitting method.

8. Challenges and future work

This section devotes to discussing the limitation of this study and how to address them in future work. An apparent limitation of this work is the lack of data. Developing a similarity-based model for RUL prediction demands the dataset of battery cell degradation from BOL to EOL. However, as mentioned before, most of the battery cells in our study did not reach EOL due to time restrictions. Due to this limitation, the curve fitting methods were adopted that impose some potential error in the process of RUL prediction. It should also be argued that some factors that can affect the process of battery PHM are neglected in this work. For instance, the proposed method only focuses on two types of NMC batteries. Moreover, the internal battery faults are skipped in this work. Considering the drawbacks of the proposed method, future work will focus on validating the proposed method using an extensive dataset that considers different types of battery chemistries and other stress factors that impact battery degradation.

9. Conclusions

The idea of prognostics and health management has attracted attention in engineering systems to ensure reliable performance and

optimize the maintenance of operating systems. This research attempts to develop a novel online data-driven methodology for prognostics and health management of lithium-ion batteries. First, a large dataset including the data of thirty-eight nickel manganese cobalt oxide cells was collected. The data are parts of two European projects which are collected over several years. The collected dataset covers a variety of stress factors that affect battery health degradation. In the next step, the health diagnosis component is developed, in which the extracted health indicators are used as input of a recurrent neural network method called nonlinear autoregressive with exogenous input (NARX) to estimate battery state of health (SOH). The results achieved from this component show that the NARX can estimate the battery SOH of unseen data with relatively low error. The estimated SOH then fitted to the health prognosis component. The curve fitting method is adopted to build a data repository of battery health trajectory in the health prognosis component. Finally, a similarity-based model was developed, which is a straightforward and precise method for predicting the remaining useful life of the battery. The accuracy of the model is validated on the battery cells that are not used for training. The findings indicate that the proposed method can predict the remaining useful life of untrained battery cells when 70% of health degradation data are available with a maximum absolute error of 1.1. In summary, the findings of this paper confirm that the proposed prognosis and health management methodology can predict the health diagnosis and prognosis of lithium batteries, with high precision, within a confidence interval and independent of their operating conditions. Therefore, it has the potential to be implemented in the battery management system in real-time operation.

CRedit authorship contribution statement

Sahar Khaleghi: Conceptualization, Methodology, Software, Validation, Investigation, Writing the original draft, Visualization. **Md Sazzad Hosen:** Resources, Experiment, Writing- Review and editing. **Danial Karimi:** Data curation, Review and editing. **Hamidreza Behi:** Data experiment, Formal analysis, Review and editing. **S. Hamidreza Beheshti:** Review and editing, Investigation. **Joeri Van Mierlo:** Supervision, Project administration, Funding acquisition. **Maitane Berecibar:** Supervision, Project administration, Review and editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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